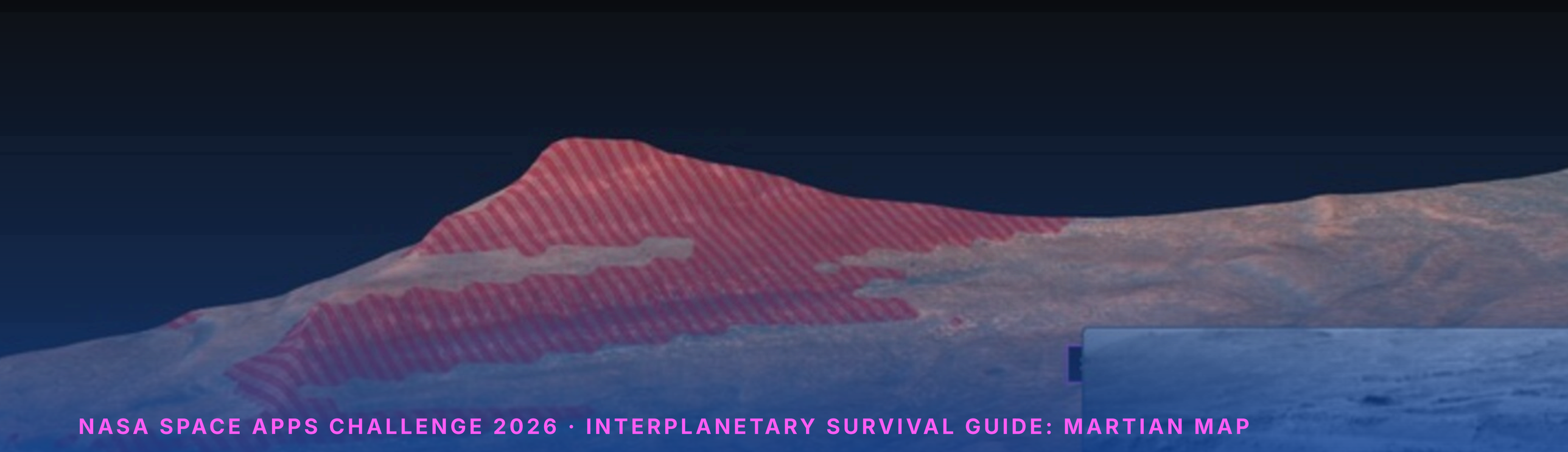


NASA SPACE APPS CHALLENGE 2026 · INTERPLANETARY SURVIVAL GUIDE: MARTIAN MAP

PROJECT TENACITY

Plan the walk on Mars before you take it. An EVA planner that sketches, simulates and validates a crew's route on real NASA data.

Team **YA RULES**
Adham Khairy, Yara Ehab

THE PROBLEM

On Mars, the limit of a walk is not distance. It is getting back to shelter in time.

60 min

is how long a suit's backup life support lasts if the main system fails. You must never be further from the refuge than that.

20°

is the steepest slope a suit is required to handle: up, down or across. A map that looks flat can hide steeper ground.

No rescue

and no satellite navigation. Oxygen, cooling water, battery, daylight and solar storms all have to be right before the airlock opens.

A crew needs one answer before they step outside: **can we go, and get home?**

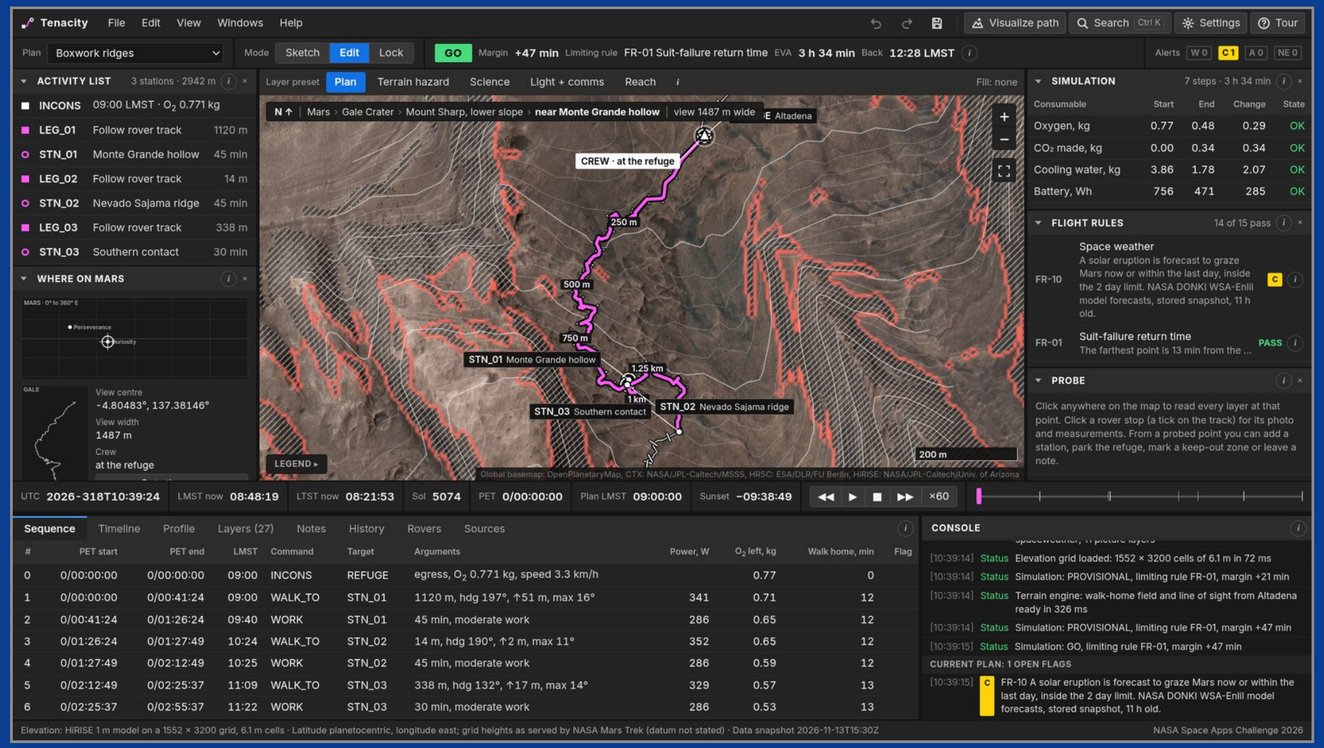
Sources: NASA xEMU life-support description (NTRS 20170009505); NASA Artemis III surface requirements (NTRS 20210026296).

THE IDEA

Rover drivers simulate every drive. Astronauts should too.

At JPL, every rover drive is sketched, simulated and checked before it is sent to Mars. Tenacity gives a crew on foot the same loop.

ROVER PLANNERS AT JPL	TENACITY, FOR A WALK
Sketch the drive on the terrain	**Sketch** legs and stations on the map
Simulate before sending	**Simulate** time, oxygen, water, battery
Check against flight rules	**Validate**: 15 rules, one GO or NO-GO



The working tool: plan on the left, map in the middle, simulation and flight rules on the right, command sequence below.

HOW IT WORKS

Click the map. The terrain engine finds the way and shows why.

1. **Sketch.** Click anywhere to add a station.
2. **Path.** A 5-million-cell elevation grid, in the browser, finds the easiest walk. **Red** cannot be crossed, **yellow** is slow.
3. **Simulate.** Pace on every slope, oxygen, water, battery, and the walk-home time from every point.
4. **Validate.** Rules name what blocks the walk and where. Missing data is "not evaluated", never a pass.



A sketched station: the dashed white line is the straight line the path finder rejected; the route goes around ground steeper than 20°.

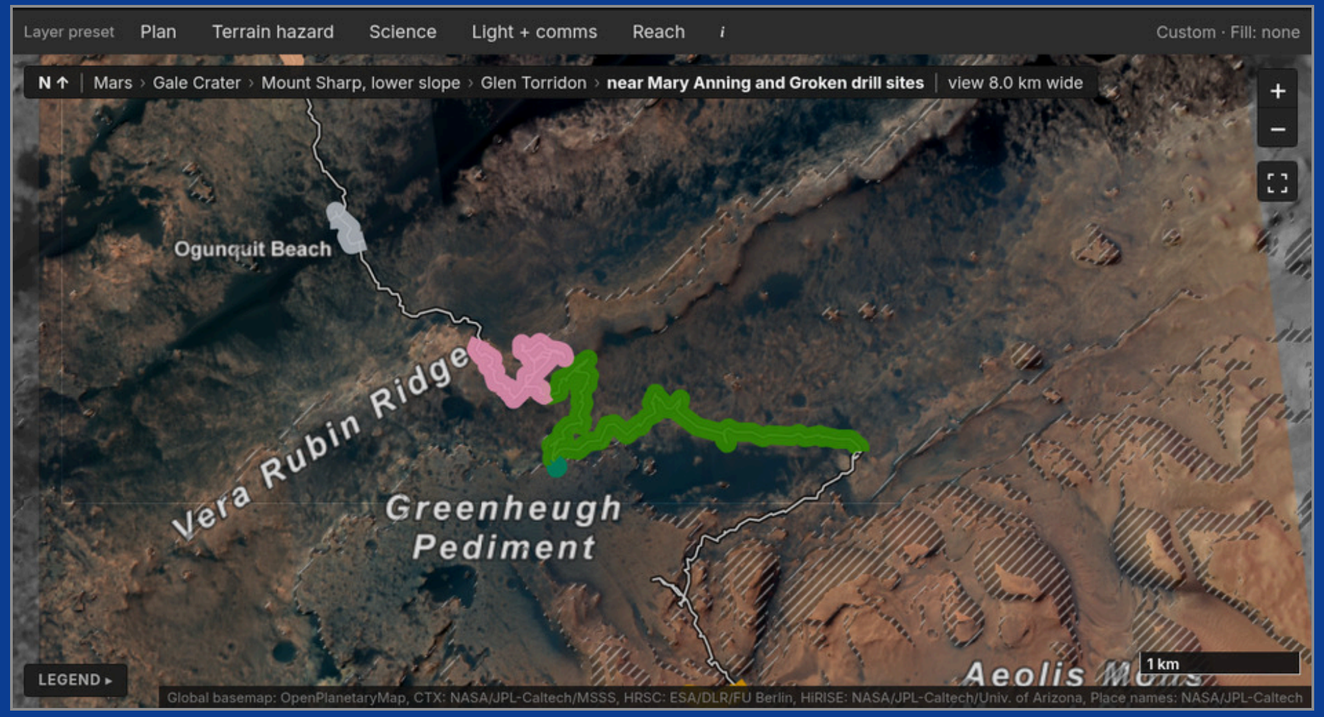
NASA DATA AT THE CORE

More than 25 layers, and every one feeds the rules.

- HiRISE imagery 25 cm
- HiRISE elevation NASA Mars Trek
- CTX imagery
- CRISM minerals
- THEMIS thermal inertia
- Curiosity track 1,388 stops
- Navcam photos
- CheMin minerals
- APXS chemistry 1,784 spots
- ChemCam targets
- DAN subsurface water
- DONKI solar-storm forecasts
- Mars24 solar time
- USGS geologic map
- Rover positions today

The data drawn on the map are the data the rules read, so the map and the plan cannot disagree.

Partner instruments on Curiosity: APXS (Canadian Space Agency), ChemCam (with CNES), DAN (Roscosmos).

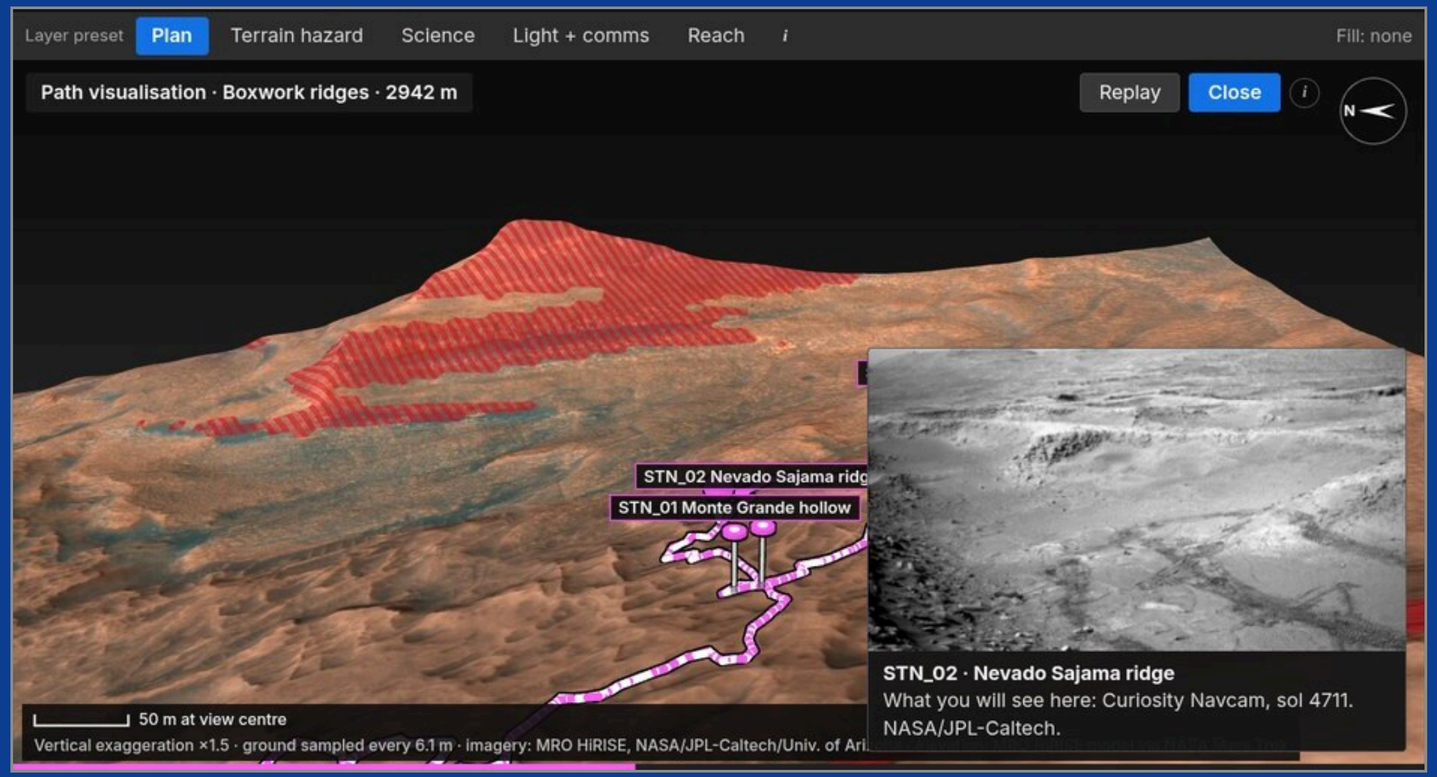


Science view: the named regions on Curiosity's route up Mount Sharp, four in the colours of NASA's own route map.

SEE IT BEFORE YOU WALK IT

A 3D flyover of the real ground, with Curiosity's own photo of where you are going.

- **Real terrain.** HiRISE elevation draped with HiRISE imagery; heights exaggerated 1.5 times and labelled so.
- **Real views.** The photo Curiosity took at each station appears as the camera arrives.
- **For the crew.** A guided tour, a large mode for gloved hands, notes on the map, and a printable cuff card.



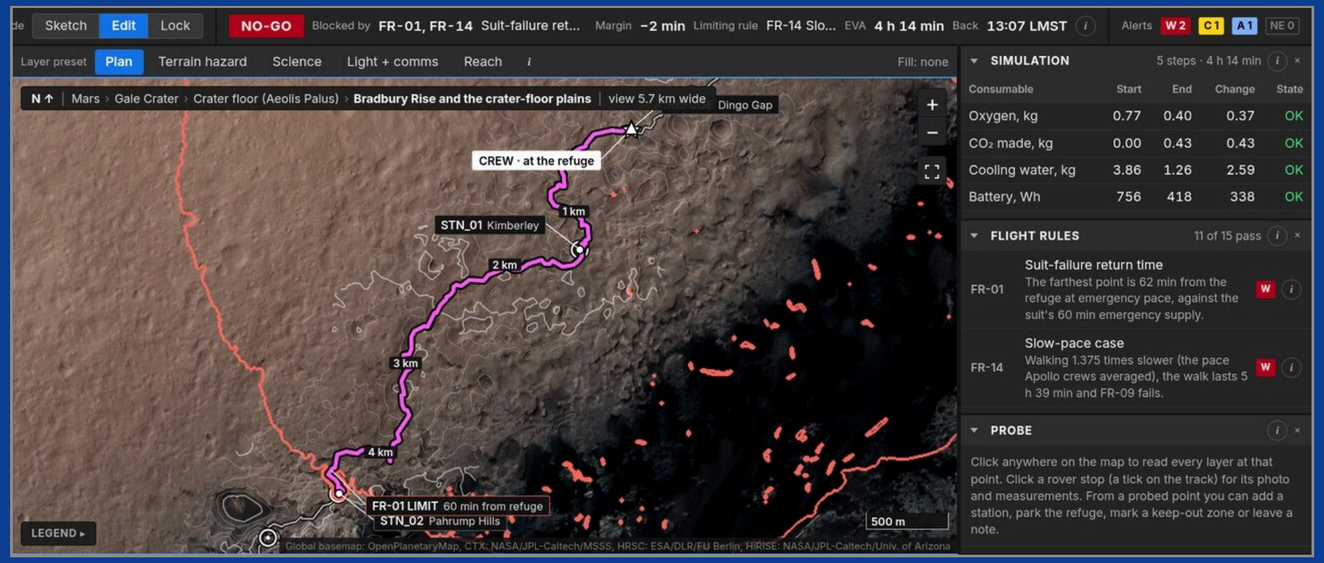
"Visualize path": the route draws itself over Mount Sharp's boxwork ridges; red hatching is ground too steep for a suit.

HONEST, OPEN, AND RUNNING NOW

A plan the tool rejects is as useful as one it clears.

- **It says no.** This 5 km walk is **NO-GO**: 62 minutes from shelter against a 60-minute limit.
- **Every number is tagged** and sourced.
  MEASURED | NASA VALUE | DERIVED | ASSUMED
- **It states its limits.** Walking speed on Mars has never been measured, and the tool says so.
- **Open source**, no server, 113 automated tests.

yara-hue24.github.io/Project-Tenacity
github.com/yara-hue24/Project-Tenacity



Next: a date for each plan, crew position tracking, and measured walking data when it exists.

Slides and much of the code were prepared with an AI assistant (Claude) under the team's direction; screenshots show the working tool. Not affiliated with NASA.